

Hojae Lee

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EDUCATION

Kyung Hee University

Seoul, KR

M.S. in Computational Health & Biomedical Data Science (School of Pharmacy) — GPA 4.25/4.5 Mar. 2023 – Feb. 2025

- Thesis: ML-based suicidal ideation prediction validated across 3 independent worldwide cohorts

University of Toronto

Toronto, CA

Applied AI Program, Dept. of Mechanical & Industrial Engineering (fully-funded MSIT/IITP fellow) Jan. 2024 – Jul. 2024

Korea University

Seoul, KR

B.S. in Mathematics

Mar. 2016 – Feb. 2023

EXPERIENCE

Researcher

Mar. 2025 – Feb. 2026

Asan Medical Center — Biomedical Engineering Research Center

Seoul, KR

- Improved QCA measurement consistency and reviewer efficiency — built and deployed analysis pipelines with clinician-facing QA interfaces
- Enabled 3D vascular geometry for TAVI planning from routine 2D angiographic sequences via 2D→3D reconstruction prototyping
- Developed semi-automated catheter/wire tracking on routine 2D angiographic sequences with MedSAM 2-assisted segmentation
- Integrated structured clinical variables with imaging model outputs for clinician-facing decision support

Independent Researcher

Mar. 2025 – Present

DEFINES LAB — Independent Research

Seoul, KR

- Extending the Toronto/MIT hallucination work to multimodal math reasoning — LangGraph multi-agent workflow with knowledge-graph structured retrieval to suppress VLM hallucination
- Implemented a self-built PyTorch OCR pipeline for document-to-structured-input conversion
- Engineered a human-in-the-loop pipeline for curriculum-aligned math problem generation and reading/grammar item QC (choice-set consistency, distractor validity, automated rationale)

Joint AI Researcher (Fully-funded MSIT/IITP Scholar)

Dec. 2023 – Jun. 2024

University of Toronto & MIT

Toronto, CA

- Improved over lightweight baselines with <1% of full fine-tuning's trainable parameters (~200k vs. 20–25M) via PEFT and soft prompting; full fine-tuning remained substantially stronger
- Boosted factual accuracy on complex scientific Q&A by embedding physics-ontology constraints through KG-RAG
- Mitigated LLM hallucination in cross-functional research teams across scientific Q&A datasets

M.S. Researcher

Jan. 2023 – Feb. 2025

Kyung Hee University — Center for Digital Health

Seoul, KR

- Authored/co-authored 20+ peer-reviewed papers within two years; built reproducible code-first reporting environments (Python/R)
- Externally validated ML risk-prediction models across two independent Korean cohorts — neurodegenerative, cardiovascular, and diabetic-retinopathy outcomes in T2D
- Standardized end-to-end clinical ML pipelines with SHAP-based XAI outputs for clinician interpretability
- Led nationwide time-series analyses of adolescent health behaviors (2005–2022), quantifying epidemiological trends and policy impacts

NLP Researcher Intern

Jun. 2021 – Jan. 2022

MEDIAZEN

Seoul, KR

- Engineered text-dataset preprocessing pipelines and structured input generation in Python for a commercial Korean chatbot system

SELECTED PUBLICATIONS

26 peer-reviewed publications | 572 citations | h-index 13 (6 co-first, 1 corresponding) — full list on [Google Scholar](#)

Long-term trends in the prevalence of cardiovascular-kidney-metabolic syndrome in South Korea, 2011–2021. *The Lancet Regional Health – Western Pacific*, 2025

ML-based prediction of neurodegenerative disease in type-2 diabetes, derived and validated in 2 independent Korean cohorts (co-first). *Journal of Medical Internet Research*, 2024

Prediction model for cardiovascular disease in patients with diabetes, derived and validated in two independent Korean cohorts (co-first). *Scientific Reports*, 2024

ML-based prediction of suicidality in adolescents with allergic rhinitis, validated in 2 independent nationwide cohorts (first author). *Journal of Medical Internet Research*, 2024

National prevalence of atopic dermatitis in Korean adolescents, 2009–2022 (co-first). *Scientific Reports*, 2024

PROJECTS

End-to-End Quantitative Coronary Angiography Pipeline | *Python, PyTorch, OpenCV, scikit-image, pytesseract* 2025 – 2024

- Built and deployed an end-to-end QCA pipeline at Asan Medical Center: sub-pixel vessel-boundary segmentation across all three coronary trees (RCA/LAD/LCX), automatic centerline and bifurcation extraction, and per-branch diameter profiling with OCR-calibrated pixel-to-mm conversion
- Checked outputs for physiological consistency against Murray's Law ($R^2 > 0.8$, $n > 1,190$ vessels) — an independent plausibility check; validation against manual QCA is the planned next step

DEFINES — Hallucination Suppression in Multimodal Math Reasoning | *LangGraph, VLM, KG retrieval* 2025 – Present

- Multi-agent VLM workflow with knowledge-graph structured retrieval for curriculum-aligned math reasoning; continuation of the UofT/MIT hallucination work

Mitigating LLM Hallucinations in Scientific Q&A | *PyTorch, HuggingFace, PEFT, KG-RAG* 2023 – 2024

- Parameter-efficient fine-tuning at <1% of full fine-tuning's trainable parameters, extended with a physics ontology-based KG-RAG pipeline grounded in Chain-of-Knowledge

Clinical Outcome Prediction at Population Scale | *Python, R, SHAP* 2023 – 2025

- Nationwide ML risk models with external validation across independent cohorts and SHAP-based interpretable outputs; published across JMIR and Scientific Reports

HONORS & AWARDS

Ministry of Science and ICT (IITP) Fellowship — Gov. of South Korea, UofT/MIT joint program Dec. 2023 – Jun. 2024

Best Paper Award — 5th Engineering in Biomedical and Rehabilitation & Home Conference, Japan Jun. 2023

AI Convergence Innovation Project Grant — Kyung Hee Univ., Chronic Disease Prediction Platform May – Dec. 2023

Kyung Hee University Academic Scholarships — Spring 2023, Fall 2023, Spring 2024 2023 – 2024

TECHNICAL SKILLS

Languages: Python, R

ML/DL: PyTorch, HuggingFace Transformers, PEFT/soft prompting, SHAP (XAI)

Medical Imaging: OpenCV, scikit-image, MedSAM 2, OCR (pytesseract), QCA/vessel analysis

LLM Systems: KG-RAG, LangGraph multi-agent workflows, hallucination evaluation